The Fibonacci-distance Neighborhood for the Permutation Flow Shop Problem

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Outline

Introduction

- **Permutation Flow Shop Problem (PFSP):** NP-hard for $m \geq 3$. Neighborhood choice drives metaheuristic quality.
- Classical neighborhoods trade off coverage and cost:
  - **Adjacent swap:** $n-1$ moves, only one swap per iteration.
  - **Dynasearch:** $O(n^2)$ candidate pairs, expensive — intractable beyond $n \sim 200$.
- **Fibonacci neighborhood:** all sets of *non-overlapping adjacent swaps* applied simultaneously — F_{n+1} valid subsets, but the optimal subset is selected by DP in $O(n)$.
- This talk: define the Fibonacci neighborhood, study its scaling behavior on Taillard benchmarks, and discuss its role within local search frameworks (ILS, SA).

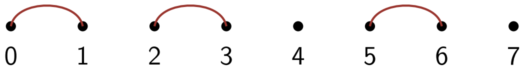
The Fibonacci Neighborhood — Definition

Definition. Given $\pi = (\pi_0, \ldots, \pi_{n-1})$, a Fibonacci move is a subset $S \subseteq \{1, \ldots, n-1\}$ of adjacent-swap positions such that

$$i \in S \implies i+1 \notin S \qquad \text{(no two consecutive swaps).}$$

All selected swaps $\pi_i \leftrightarrow \pi_{i+1}$ are applied *simultaneously*.

Example (admissible: swap at 0, 2, 5; positions 1, 3, 4, 6 idle):



Counting. The number of valid subsets in a permutation of length n equals the Fibonacci number F_{n+1} — hence the name.

Properties and Computation

- **Exponential size, linear cost.** $|N_{\mathrm{fib}}(\pi)| = F_{n+1} \sim \varphi^n/\sqrt{5}$, yet the optimal subset is found in $O(n)$.
- **Algorithm:** compute the $n-1$ adjacent-swap deltas using Head+Tail in $O(mn)$, then a one-pass DP over the deltas:

$$\mathrm{dp}[i] \;=\; \min\big(\,\mathrm{dp}[i+1],\; \delta_i + \mathrm{dp}[i+2]\,\big)$$

  decides at each position whether to take the swap or skip it.
- **Acceleration:** the Smutnicki block-property and the NPI filter prune dominated swaps before the DP — typical reduction 30–60%.
- **Compared with Dynasearch.** Same compound-move idea, but Fibonacci only considers adjacent pairs $(i, i+1)$ while Dynasearch considers arbitrary (i, j) — $n-1$ candidates vs. $O(n^2)$.

QUBO Formulation for D-Wave

Binary variable $x_i \in \{0, 1\}$ for each adjacent-swap position $i \in \{1, \ldots, n-1\}$: $x_i = 1$ iff swap $\pi_i \leftrightarrow \pi_{i+1}$ is applied.

QUBO form — tridiagonal coupling:

$$Q_{\mathrm{fib}}(x) = \sum_{i=1}^{n-1} \delta_i \, x_i + P \sum_{i=1}^{n-2} x_i \, x_{i+1},$$

where δ_i is the precomputed adjacent-swap cost and $P \gg \max |\delta_i|$ penalizes overlapping swaps (encoding the non-overlapping constraint).

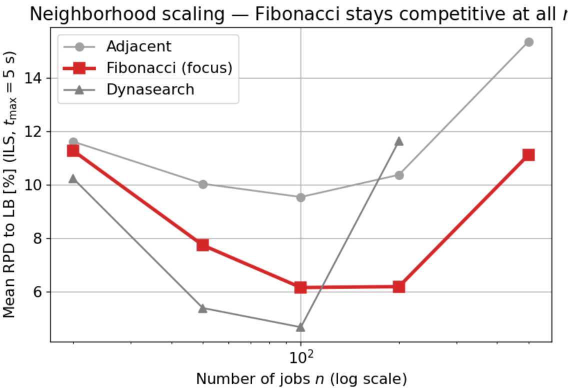
On D-Wave Advantage:

- Bandwidth 2 $\Rightarrow$ linear-chain embedding on Pegasus/Zephyr, no chain breaks.
- Logical qubits $= n-1$; feasible up to $n \leq 500$ on current hardware.

Experimental Setup

- **Benchmarks:** Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$, 10 instances each, 5 seeds.
- **Time limits:** $t_{\max} \in \{0.1, 0.5, 1, 2, 5, 10\}$ s; metric: mean RPD to lower bound.
- **Algorithms:** Iterated Local Search (tabu τ=10, MushroomList k=10); Simulated Annealing (T_0=1000, α=0.99, reheat-kick on stagnation).
- **Quantum:** D-Wave Advantage 4.1, 5,000 reads, windowed decomposition for $n > 19$.
- **Compared neighborhoods:** Adjacent, **Fibonacci**, Dynasearch.

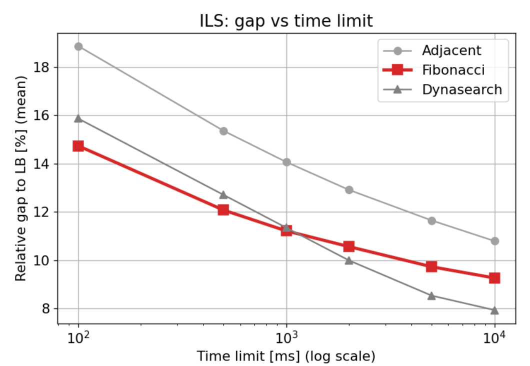
Scaling Behavior — RPD vs. n

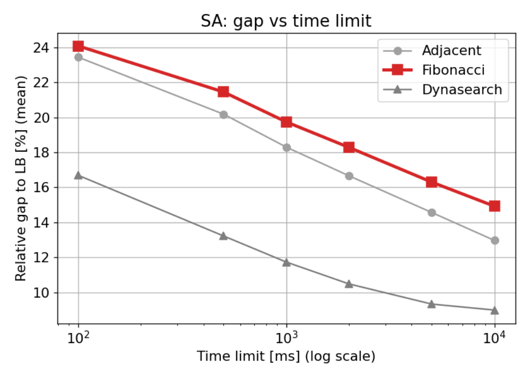


Observations (ILS, $t_{\max} = 5\,\mathrm{s}$, m=10):

- Up to $n \approx 100$: Dynasearch wins, Fibonacci second.
- $n = 200$: Fibonacci **overtakes** Dynasearch (6.19 % vs. 11.63 %).
- $n = 500$: Dynasearch becomes intractable; Fibonacci continues at **11.12 %** — the only classical neighborhood that still scales.

Gap vs. Time Limit





- Fibonacci has the flattest curve — already near its plateau at $t_{\max} = 100$ ms.
- Dynasearch reaches lower RPD at small n but needs the full 10 s budget to get there.

Conclusion and Future Work

Summary:

- Fibonacci selects an optimal set of non-overlapping adjacent swaps applied simultaneously — a multi-swap move at distance 1.
- Exponential neighborhood size (F_{n+1}) explored at linear $O(n)$ cost via dynamic programming.
- Tridiagonal QUBO embeds cleanly on D-Wave Advantage; feasible up to $n = 500$.
- Empirically beats Dynasearch from $n \geq 200$, where Dynasearch's $O(n^2)$ candidate enumeration runs out of budget.

Future work:

- Full QPU evaluation of the Fibonacci QUBO at $n = 500$ on D-Wave Advantage.
- Relaxing the non-overlapping constraint — e.g. minimum-gap variants.
- Combining Fibonacci selection with cyclic / random subset perturbation as a diversification operator.